



Awesense Energy Transition Platform Use Cases



- demoable implementation



- demoable & open sourced implementation

Feb 7, 2024



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Power BI



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UC01 Coincident Peak and Non-Coincident Peak Analysis



UC02 Power Factor Analysis



UC03 Phase Balance Analysis



UC04 Transformer Capacity Analysis for EV Charging



UC05 Reversed Power Flow and PV Capacity Analysis



UC06 GIS Grid Connectivity Validation and Correction



UC07 Asset Management - Cables and Lines

UC08 Asset Management - Substations

UC09 EV Charging and Use of Reserved Capacity Analysis



UC10 Connected Loads [kW] vs. Actual Phase Balance Analysis

UC11 Analysis of Unserved Energy Due to Outage Analysis

UC12 Grid Temperature Monitoring



UC13 Master Circuit Breaker Value vs. Measured Values Analysis



UC14 Onsite Generation and Ghost Load Analysis

UC15 Utility Data Challenge Hub



UC16 Registered PV Generation Analysis



UC17 Actual vs. Contracted Grid Capacity Analysis



UC18 Grid Alerts and Warnings



UC19 IoT Health Index and Overview



UC20 Unexpected Generation Identification

UC21 Geospatial Analysis of Grid Alerts and Warnings



UC22 Topological Grid Segmentation





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UC23 Analysis of Grid Segments and Elements



UC24 Analysis of Planned Outage for Assets Upgrades



UC25 Analysis for Maximizing Value of IoT and Meter Devices



UC26 Transformer Loading Analysis



UC27 IoT Sensor Billing



UC28 Energy Balancing for Revenue Protection



UC29 Non-Technical Loss Investigations Management



UC30 Suspicious Pattern Identification & Analysis



UC31 Grid and Time Series Visualization



UC32 Power Outage Visualization



UC33 MV Grid Optimization Analysis

UC34 EV and Appliance Disaggregation

UC35 Standard Consumption Patterns Creation

UC36 Phase Identification Analysis

UC37 Detection of Meter Incorrect Wiring Connection

UC38 Transformer Load Increase Assessment Tool

UC39 IPT and ICT Verification for Proper Billing



UC40 Phase Imbalance Correction Suggester

UC41 EV Load Growth Forecasting



UC42 Meter to Transformer Association Analysis

UC43 Load Forecasting

UC44 Voltage Power Quality Monitoring





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UC45 Feeder Voltage Analysis and Visualisation

UC46 Feeder Capacity and Lateral Capacity Analysis

UC47 Dynamic Line Rating Analysis

UC48 AMI-Smart Meter Flag Analysis for Theft Detection



UC49 Time Of Use Rate Analysis

UC50 Work Orders Management



UC51 Wildfire Risk Assessment